MATH290

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Proposition 1. *For any integer k, $k^3 - k$ is a multiple of 6.*

Proof.

Firstly, recall that an integer is a multiple of an integer 6 if it can be expressed as the product of 6 and some other integer. Now, observe what happens if you substitute $-k$ for k:

$$\begin{aligned}&(-k)^3 - (-k)\\ =&-(k^3) + k\\ =&-(k^3 - k)\end{aligned}$$

Let f_k be the bi-infinite sequence $k^3 - k$ for all integers k. It turns out that rather than proving each term in this sequence is a multiple of 6, it's easier to prove that *the difference between consecutive terms* is a multiple of 6. (Or at least *I* found it to be.) Let's let g_k be the difference between any f_{k+1} and f_k. That is,

$$\begin{aligned}g_k &= f_{k+1} - f_k\\ f_{k+1} &= f_k + g_k\end{aligned}$$

Consider then that the sum of two multiples of 6 is itself a multiple of 6. I'll demonstrate this for two multiples of six $n = 6i$ and $m = 6j$, where n, m, i, and j are all integers. Observe,

$$\begin{aligned}n + m &= 6i + 6j\\ &= 6(i + j)\end{aligned}$$

Relying on the property that the sum of integers is an integer, we've shown that $n + m$ can be expressed as a product of 6 and some other integer $i + j$, and thus $n + m$ is a multiple of 6.

Thus, if I can show that every term in g is a multiple of 6, I can use mathematical induction to show that every term in f is the sum of two multiples of 6—in which case all terms of f are themselves multiples of 6.

To start, I'll show that g_k is a multiple of 6 for all integers k. Because 6 is equal to 2×3, I can prove g_k is a multiple of 6 by proving it is a multiple of both 2 and 3 for any integer k.

I can prove g_k is always a multiple of 3 by showing it can be expressed as the product of 3 and some other integer. Observe,

$$\begin{aligned} g_k &= f_{k+1} - f_k \\ &= (k+1)^3 - (k+1) - (k^3 - k) \\ &= (k+1)\big((k+1)^2 - 1\big) - (k^3 - k) \\ &= (k+1)(k^2 + 2k) - (k^3 - k) \\ &= k^3 + 3k^2 + 2k - k^3 + k \\ &= 3k^2 + 3k \\ &= 3(k^2 + k) \end{aligned}$$

To prove that g_k is a multiple of 2 for all integers k, I will prove that g_k is *even* for all integers k. Firstly, recognize that for any integer k, if $k^2 + k$ is even, then $3(k^2 + k)$ is even. Then, recognize that for any *even* integer k, both k^2 and k are even, and so their sum $k^2 + k$ is even. Lastly, recognize that for any *odd* integer k, both k^2 and k are odd, and so $k^2 + k$ is even. So, regardless of whether an integer k is even or odd, $k^2 + k$ is even and therefore $g_k = 3(k^2 - k)$ is also even.

(This line of reasoning relies on the knowledge that the product of an even and an odd is even; the square of an even is even; the square of an odd is odd; the sum (or difference) of two evens is even; and the sum (or difference) of two odds is even.)

Therefore, having shown that g_k is a multiple of 3 and even for every integer k, I have proven that g_k is a multiple of 6 for every integer k.

The rest can be proven by mathematical induction on k. Suppose that for some integer k, f_k is a multiple of 6. Consider then the case for $k + 1$. Because f_k and g_k are both multiples of 6, their sum, f_{k+1}, is also a multiple of 6. Thus, the result holds for the inductive case.

Finally, consider the base case f_0. Observe,

$$\begin{aligned} f_0 &= (0)^3 - (0) \\ &= 0 \end{aligned}$$

0 is a multiple of 6 (it can be expressed as 6×0). Thus, the result holds for the base case $k = 0$. Then, by the principle of mathematical induction, $k^3 - k$ is a multiple of 6 for every integer k.

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